# Project Phase 4: Final Report

CMPT 276: Introduction to Software Engineering
Simon Fraser University

**Group #15**

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## 1 The Game

### 1.1 Game Description

*Meowmino's Delivery* is a 2D tile-based side-scrolling delivery game built with LibGDX (LWJGL3 desktop backend). The player controls a cat riding a delivery bike across three connected maps. The core gameplay loop is: collect fish scattered on road tiles, watch for timed delivery orders from houses, dismount the bike to walk up to a house and deliver, then remount and continue — all before a 5-minute countdown expires. Two hazard types oppose the player: **police enemies** that patrol the maps and give chase on sight (respawning the player with a fish and time penalty if caught), and **puddle hazards** that slow movement and continuously drain collected fish while the player overlaps them. Progress through the three maps is locked behind gates: Gate 1 requires 30 fish; Gate 2 requires at least one delivery to every house on Map 2. A special ending triggers on Map 3 when the player reaches the White House landmark with 90 or more fish, playing a full animated cutscene.

### 1.2 Fidelity to Original Plan

The following core features from the Phase 1 plan were delivered as designed:

- Three-map tile world ($5446 \times 2706$ px total) with pixel-accurate barrier-based collision
- Cat-on-bike player character with WASD movement and directional animation
- Fish collectibles (20 per map, 60 total) randomly spawned on road tiles

- Police enemies with configurable detection radius and state-machine chase AI
- Delivery houses with timed orders and fish rewards (+10 per delivery)
- Gate-based map progression gated by fish quota (Gate 1) and delivery history (Gate 2)
- Five-minute countdown timer ending the game on expiry
- EndScreen displaying a zero-padded score and elapsed time using digit sprites
- Pause overlay with Resume, Restart, and Quit options
- Settings screen with persistent sound on/off toggle

## 1.3 Changes from the Original Plan

Several features were added or modified during development. The table below summarises each change and its justification.

| **Feature** | **Status** | **Justification** |
|---|---|---|
| Mount/dismount mechanic (Q key; `CatOffBike` entity) | Added | Adds strategic depth: the player must physically walk to a door, creating risk vs. speed trade-offs |
| Off-bike requirement for delivery | Added | Natural complement to mount/dismount; incentivises route planning |
| Obama / White House cutscene ending | Added | Narrative payoff for completing all three maps with 90+ fish; rewards thorough play |
| `DeliveryNotification` ticket UI (slide-in tickets, timer bar, 3 slots) | Added | Phase 1 showed a basic indicator; the full ticket system gives clear, real-time order feedback |
| Iris wipe transition + cinematic letterbox bars | Added | Polish: smooth map transitions and framed cutscenes improve perceived game quality |
| Map 3 camera pan cinematic on entry | Added | Previews house locations before gameplay resumes, reducing navigation confusion |
| Gate 2 condition changed to “all houses delivered” | Changed | Fish quota alone gave no delivery incentive on Map 2; requiring deliveries ties gate logic to the game’s core mechanic |
| Puddle fish-drain effect (in addition to slow) | Added | A single slow effect felt minor; ongoing fish drain makes puddles a meaningful hazard |

### 1.4 Lessons Learned

**Testing LibGDX entities requires upfront infrastructure planning.** Entity constructors load textures and audio, requiring an active OpenGL context. We solved this with Objenesis (to instantiate entities without calling constructors) and Mockito (to stub LibGDX singletons such as `Gdx.graphics` and `Gdx.audio`), but building this infrastructure mid-project was costly. Future projects should design and validate the test harness before writing entity code.

**Separating pure logic from rendering pays dividends throughout development.** Helper classes `GameLogicHelper`, `ObamaDialogueHelper`, `GateChecker`, and `Respawn` contain no LibGDX dependencies and are trivially unit-testable. Identifying and extracting this logic early would have saved significant refactoring effort that was instead performed after the fact during the testing phase.

**Peer code review reveals blind spots that self-review misses.** Assignment 4 code reviews identified concrete smells: duplicated barrier-collision logic in `PoliceEnemy`, data clumps for spawn coordinates in `GameScreen`, an unnecessary `bikeParked` variable mirroring `isOnBike`, and dead code. Addressing this feedback improved both readability and test reliability.

**Post-implementation testing is significantly more expensive than TDD.** Most tests were written after features were implemented. Retrofitting testing constructors, extracting helpers, and writing meaningful assertions took as long as the tests themselves. The lesson is clear: design for testability from day one.

## 2 Tutorial / Demo

### 2.1 Video Demonstration

A full gameplay demonstration is available at:
`https://youtu.be/gFfb3DsHz78?si=UlMbHLLywqt6zSot`
The video covers all major mechanics: fish collection, delivery orders, police and puddle hazards, gate progression through all three maps, and the special Obama ending cutscene.

### 2.2 How to Play

**Controls**

| **Key** | **Action** |
|---|---|
| W / A / S / D | Move up / left / down / right |
| Q | Mount or dismount the bike |
| E | Deliver an order (off-bike, within 150 px of a house with an active order) |
| ESC | Pause / unpause |

**Objective.** Collect fish on road tiles, fulfil delivery orders from houses, and advance through three maps before the 5-minute timer reaches zero. Reaching the White House on Map 3 with 90 or more fish triggers the special ending.

**Fish Collection.** Fish icons are scattered across the road. Walk (or ride) over a fish to collect it. Each fish increments the counter shown in the HUD. Deliveries award a bonus of +10 fish per completed order.

**Delivery System.** When a house has an active order, a delivery ticket slides onto the screen showing the house and a countdown timer. Dismount the bike with Q, walk to the house, and press E to deliver. Orders expire after 10 seconds. Up to three orders can be active simultaneously.

**Hazards.** *Police enemies* patrol each map. When a police officer spots the player (within 400 px), a question mark appears above their head followed by a chase. Being caught respawns the player at the map start with −10 fish and −30 seconds. *Puddles* slow movement speed from 400 px/s to 70 px/s and drain 1 fish per frame while the player overlaps them. Avoid or cross quickly.

**Gate Progression.** Gate 1 (Map 1 → Map 2): collect 30 or more fish. Gate 2 (Map 2 → Map 3): deliver to every house on Map 2 at least once.

**Special Ending.** On Map 3, navigate to the White House landmark with 90 or more fish collected to trigger the Obama cutscene and complete the game.

# 3 Build Automation

All build, test, and artifact tasks are driven by Apache Maven 3.6+ with Java 11. No additional IDE or toolchain setup is required.

| **Task** | **Command** |
|---|---|
| Compile sources | `mvn clean compile` |
| Run the game | `mvn clean compile exec:exec` |
| Run all tests | `mvn test` |
| Run tests + JaCoCo coverage report | `mvn test jacoco:report` |
| Package executable shaded JAR | `mvn clean package` |
| Run the shaded JAR | `java -jar target/java-game-1.0-SNAPSHOT-shaded.jar` |
| Generate Javadoc | `mvn javadoc:javadoc` |

Coverage reports are written to `target/site/jacoco/index.html`. Javadoc output is written to the `javadoc/` directory at the project root.

**macOS note:** the shaded JAR requires an extra JVM flag due to LWJGL3:

```
java -XstartOnFirstThread -jar target/java-game-1.0-SNAPSHOT-shaded.jar
```